Lab 6: bench checklist

Distances

Use the course LED lamp at fixed power. Measure from its light-emitting front surface to the nearest glass surface at rotor height: **2 cm, 4 cm, 6 cm, then repeat 2 cm after cooling**. Keep background **25 cm behind the rear glass surface** and out of the strongest beam. Measure actual lux beside the bulb at each distance.

Camera

1280 x 1024 pixels; 1000 capture frames/s. Other settings are preset. Check a short movie, verify p over one full turn and confirm that the available recording duration captures at least seven passage marks.

Record

Start from a cooled bulb and rotor at rest. Start the recording before light-on. Save 2cm_1.mp4, 4cm.mp4, 6cm.mp4 and 2cm_repeat.mp4 as H.264 with Text Overlay off. Keep distance, lux, capture fps, p and light-on frame with the movies.

Analyze

Mark_Lab6_Passage_Frames.m saves passage CSVs. Run Run_Lab6_Radiometer.m for each. Complete the prefilled Lab6_Comparison_Log.csv and run Compare_Lab6_Illumination_Runs.m. Use the same supported physical-time window for all comparisons.

Keep

Raw movies, passage CSVs, completed comparison log and exported figures. The main two-page guide lists the report figures and the required Q6 connection.